

Sebastian Cuaresma

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EDUCATION

University of Maryland

Bachelor of Science

Major in Mathematics, Statistics Track; Minors in Business and Economics

Cumulative GPA: 3.64/4.0; Academic Honors 2024–2025

Relevant Coursework: Linear Algebra, Probability Theory & Statistics, Statistical Methods, Algorithms, Machine Learning, Micro & Macro Economics, Financial Forecasting, Supply Chain Management

College Park, MD

Expected May 2028

WORK EXPERIENCE

National Security Agency (Rotational Program, Part-Time)

Fort Meade, MD

Machine Learning Engineer

Jul 2026 – Present

- Architected and implemented GitLab CI/CD pipeline using Kaniko for automated Docker image builds of MCP server system, replacing manual processes on individual workspaces
- Integrated enterprise AI agent with 4 organizational tool servers deploying 29 automated capabilities; resolved authentication/transport compatibility issues, automated deployment workflow reducing setup time by 80% through dynamic user detection and prerequisite validation, and documented troubleshooting procedures
- Established LLM tool-calling evaluation framework comparing GPT-5.4 and Gemma 4 31B against Claude 4.5 Sonnet baseline using 100-point scoring methodology across 6 categories; designed 16 standardized test prompts validated against production MCP tools and delivered formal comparative analysis report with deployment recommendations
- Built two production-grade Python tools for enterprise tool servers enabling resource retrieval and software inventory management; established separate validation workflows meeting linting standards, and successfully migrated codebase from development to production repository

AI Researcher

Jul 2025 – Jul 2026

- Architected comprehensive LLM evaluation framework for quantization analysis across GPT, DeepSeek, and Gemma model families; developed Python pipeline using ROUGE-1/2/L metrics with composite weighting, conducted statistical analysis across 50+ adversarial test prompts spanning 6 categories using ANOVA and performed refusal behavior analysis identifying performance divergence
- Developed AI authorship attribution model in Python using the difflib library to distinguish between human and AI-generated code across three LLM architectures; optimized SVM classifier by adjusting embedding windows, developed multi-dimensional visualization framework, using scatter plots and similarity heatmaps, and improved model-specific identification accuracy for similar architecture families
- Built linear regression model in Python achieving 86% explained variance for severity scoring framework; applied bootstrapping to expand dataset representativeness and developed decision tree models for feature importance ranking with statistical significance testing

Data Scientist

Jul 2024 – Jul 2025

- Developed RAG optimization model in Python creating contextual text generation for semantic metadata files improving retrieval accuracy; designed meta-prompt architectures for code generation, and researched OCR preprocessing integration for image-based documents
- Built XGBoost forecasting model in Python achieving best-in-team performance predicting \$1B+ budget disbursements with hundreds-level precision; developed SARIMA model with seasonal decomposition after ARIMA showed high residual variance through ACF/PACF analysis
- Designed and delivered ML training program for 50+ non-technical professionals covering Python fundamentals, pandas, visualization, and ML concepts; created 10+ interactive Jupyter notebooks with 200+ views and 85% completion rate

National Security Agency (High School Work Study Program)

Fort Meade, MD

Data Analyst Intern

Sep 2023 – Jul 2024

- Built K-means clustering model in Python on 1M+ row transportation dataset using elbow method and silhouette analysis revealing 5 distinct usage patterns; conducted exploratory data analysis on HR metrics identifying seasonal trends with actionable recommendations and developed onboarding documentation reducing new user ramp-up time by 40%

UNIVERSITY PROJECTS

Automated Investment Portfolio

Sep 2026

- Built a formula-driven Excel model backtesting a \$500,000, 12-stock IT and healthcare portfolio (Jan 2020 – Jan 2025) under diversification rules (3–20% per stock, 30% sector minimum, 5% cash cap); calculated shares, tiered transaction fees, 15% tax, and net gain for a 541.6% pre-tax return
- Created industry and sector summary tables and a 3-year rebalancing simulation (sold the 6 weakest performers, reinvested in the top 6) that beat buy-and-hold by \$279,679 (11.6%); documented methodology and findings

Research Report: Data Ownership

Jan 2026 – May 2026

- Co-authored a research report arguing that users should hold legal ownership of the data they generate online; wrote a literature review of academic, legal, and policy sources on surveillance capitalism, data breaches, doxing, and ad-funded “free” services

Credit Card Customer Segmentation Using K-Means Clustering

May 2025

- Built K-Means clustering model in Python (scikit-learn, pandas) on ~10,000 bank credit card customers using 19 encoded and standardized demographic and behavioral features; applied elbow method ($K = 1-10$) to select $K = 5$ and visualized segments with PCA (matplotlib, seaborn)
- Profiled the five segments on credit limit, transaction volume, utilization, and retention (power users: \$12.4K average spend, 96% retained; basic-card segment: 67% retained, the highest churn)
- Developed segment-specific recommendations (reward upgrades, cashback, priority retention, credit-building options) and presented findings in a slide deck

ACTIVITIES

Web Committee (Rotational Program, Part-Time) President / Committee Lead

Fort Meade, MD

Mar 2026 – Present

- Lead team of 5 developers in maintaining internal web infrastructure; coordinate monthly sprints and conduct individual technical mentorship sessions
- Spearheaded internal program site modernization project: updated 200+ entries, restructured navigation architecture, and standardized metadata schemas
- Maintain 3+ databases for program including web site, internal intern boards, rosters, etc.

AI/ML at UMD

College Park, MD

Club Member

Apr 2025 – Present

- Collaborate on research and personal projects exploring transformer architectures, reinforcement learning algorithms, and neural network optimization techniques
- Attend and present at technical workshops on prompt engineering and RAG system design to 30+ club members

MLcon & SANER Conferences

Richmond, VA

Presenter

Oct 2025

- Selected to present machine learning research on AI exploit severity classification at two major IC technical conferences (MLcon and SANER) attended by 50+ data scientists, researchers, and intelligence analysts
- Delivered technical briefing on predictive modeling methodology including feature engineering approach, bootstrapping techniques for bias mitigation, and linear regression model achieving 86% explained variance

ADDITIONAL

Python & Machine Learning: Advanced proficiency in Python (pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers, matplotlib, seaborn, Jupyter); supervised and unsupervised learning, feature engineering, model evaluation, LLM evaluation, RAG

Modeling & Statistics: K-Means clustering, PCA, linear regression, decision trees, SVM, XGBoost, ARIMA/SARIMA time series forecasting, bootstrapping, ANOVA, statistical significance testing

Tools & Other: MATLAB, SAS, GitLab, CI/CD (Docker, Kaniko), Linux, Claude Code, Coder (cloud development environments), Data Visualization, Geospatial Mapping, Technical Writing

Certifications & Training: National Cryptologic University (NCU) Courses: Data Literacy Fundamentals, Data Science Foundations, AI & GenAI Fundamentals

Honors & Recognition:

- **Project Technical Lead** – Led 4 concurrent initiatives: LLM Evaluation Framework, ML Training Infrastructure, Classification Modeling, Time Series Forecasting
- **Performance Awards** – 7 awards for technical excellence and project delivery through work initiatives (2024–2026)